Climate Impact and Correlation Analyzer

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Abstract— Climate change analysis demands integrating and exploring multidimensional data encompassing surface temperatures, carbon emissions, sea-level rise, and natural disaster frequencies. We present an interactive visualization platform that correlates multiple climate indicators over space and time, enabling users to uncover complex, non-linear relationships. Using Python-based analysis and visualization libraries (e.g., Plotly, Chart.js) and climate-related datasets from Kaggle and other sources, we developed a suite of interactive charts—choropleth maps, heatmaps, bar charts, line graphs, 3D globe visualizations—to reveal trends and potential causal linkages. Our tool highlights associations between rising emissions, increasing temperatures, sea-level changes, and enhanced disaster frequency. The resulting insights aim to inform policymakers, researchers, and the general public, supporting evidence-based decision-making and future risk forecasting.

1 Introduction

The global climate crisis has manifested in rising temperatures, melting ice caps, and shifting precipitation patterns, driving disruptions in ecosystems, agriculture, and human health. Increasing greenhouse gas emissions, notably CO_2, have contributed to intensified climate variability, while sea-level rise and an uptick in natural disasters underscore the urgency of understanding these interconnected phenomena. Yet, conventional climate data tools often struggle to reveal the complex, multidimensional relationships that underlie these changes.

To address this challenge, we developed an interactive climate data visualization platform that integrates multiple datasets—surface temperature, carbon emissions, sea-level rise, and natural disasters—into a single analytical environment. Our approach leverages Python-based analysis (e.g., Pandas, NumPy) and visualization libraries (Plotly, Chart.js) to generate dynamic, user-friendly interfaces. By enabling users to filter data by country, region, and time, and by providing multiple representation formats (e.g., heatmaps, choropleth maps, 3D globes), the platform fosters a nuanced understanding of climate change dynamics.

This integrated perspective helps highlight how emissions correlate with temperature increases, how these temperature anomalies relate to sea-level trends, and how both temperature and emissions patterns may be associated with natural disaster occurrences. Ultimately, our work supports more informed decision-making, guiding researchers, policymakers, and other stakeholders toward more effective climate adaptation and mitigation strategies.

2 Related work

Previous climate data visualization efforts include static maps, charts, and model outputs, such as IPCC summaries [1] and NASA's GIS-based temperature anomaly maps. Other tools have provided interactive dashboards with single-variable focuses—e.g., emissions or temperature only—limiting their ability to uncover subtle relationships. Liu et al. [2] introduced an emissions-to-temperature dashboard, while Smith et al. [3] emphasized connections between emissions and sea-level changes. Fewer works integrate natural disaster records into the analytical framework, potentially missing key signals about how warming affects disaster frequency.

In contrast, our platform builds on these studies by unifying multiple climate indicators in a single environment. By doing so, we offer a comprehensive lens on climate dynamics, allowing users to explore causal and correlational linkages that might not emerge from isolated datasets. The additional dimension of natural disaster data, spanning various event types (floods, droughts, extreme weather), provides essential context for understanding the societal and environmental costs of climate shifts.

3 Datasets and pre-processing

Our visualization platform integrates multiple integrates temporal and spatial climate-related datasets sourced from Kaggle and other publicly available archives. These datasets include:

- **Global Surface Temperature**— Monthly average surface temperatures from the 18th century to present, enabling long-term trend analysis of warming patterns.
- **Carbon Emissions**—Country-level carbon dioxide equivalent emissions spanning from the early 20th century through recent decades, allowing the examination of historical emission trajectories.
- **Sea Level Rise**—Historical mean sea level (MSL) records extending back to the 19th century, capturing long-term changes associated with glacial melt and thermal expansion.
- **Natural Disasters**—A compilation of events (e.g., floods, droughts, earthquakes, extreme weather) from the early 20th century to the present, permitting correlation of disaster occurrences with temperature anomalies and emissions trends.

- **Data Cleaning and Transformation:** We removed irrelevant columns, handled missing values through interpolation or averaging, and standardized units. For example, temperature anomalies were consistently expressed in degrees Celsius, emissions in metric tons of CO_2-equivalent, and sea levels relative to a baseline MSL. By aligning temporal and spatial granularity—often converting data into a consistent "Country–Year" format—we facilitated cross-dataset comparisons.
- **Exploratory Data Analysis (EDA):** Before creating visualizations, we employed statistical techniques (e.g., calculating averages, medians, correlation coefficients) and produced preliminary plots to guide the selection of final visualizations. This EDA step informed decisions on which datasets to correlate and how best to highlight significant patterns.

4 IMPLEMENTATION

The Fig. 1 is the Homepage of our interface. The page in Fig. 1 consists of tabs that lets us select and view the visualizations for Climate Indicators and Correlations.

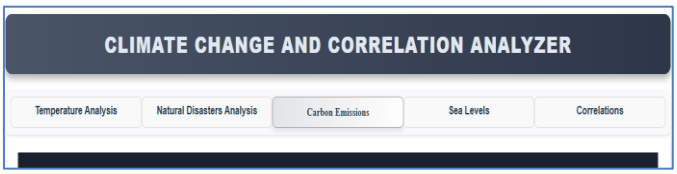

Fig. 1. Home Page

4.1 Temperature Change

Calendar Heat Maps: Showing daily temperature variations and seasonal patterns for a selected year and location (Fig 2).

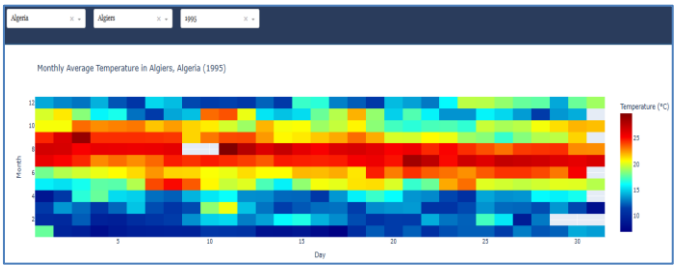

Fig. 2. illustrates a calendar heat map for a given city-year combination

3D Globe Representation: A global perspective on temperature disparities, allowing users to rotate, zoom, and inspect different regions (Fig 3).

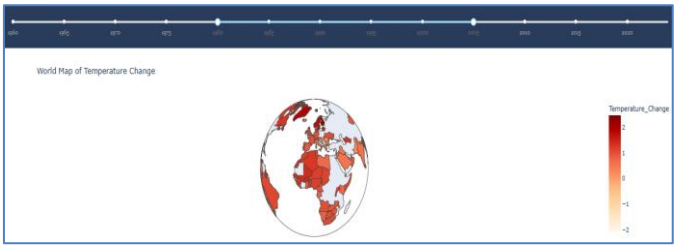

Fig. 3. Represents a 3D globe of temperature anomalies with time sliders

Choropleth Maps: Displaying spatial patterns of temperature changes over time with adjustable temporal slides (Fig 4).

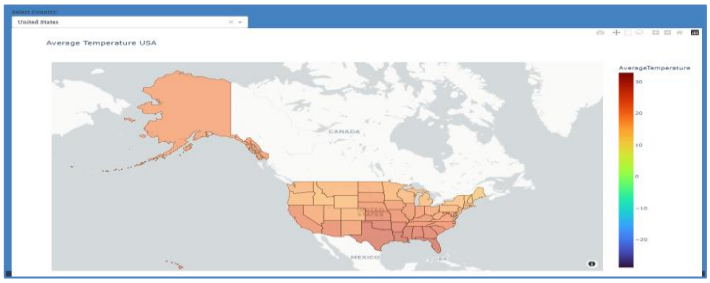

Fig. 4. Choropleth Map

4.2 Carbon Emissions

Heat Maps & Bar Charts: Highlighting regional and temporal variations in emissions, identifying top emitters, and tracing emission trajectories over decades (Fig 5, Fig 6, Fig 7).

Line Charts: Tracking how both leading and lagging emitters evolve, clarifying whether global mitigation efforts are succeeding.

Fig. 5. Global heat map showing average temperature changes by country, highlighting regional variations

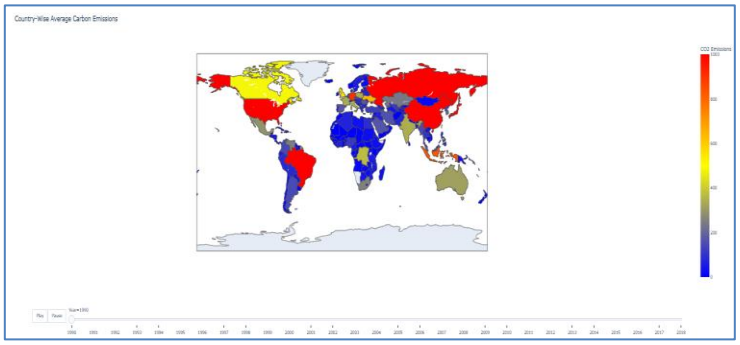

Fig. 6. Country-wise average carbon emissions displayed on a global map, highlighting regional disparities.

Fig. 7. Carbon emissions by country over time, with top-emitting countries' growth

4.3 Sea Level Rise

Bar, Scatter, and Box-and-Whisker Charts: Offering complementary views of sea-level changes from 1880 to 2020, helping to understand both the magnitude and variability of rise (**Fig 8**).

Fig. 8. Sea level change from 1880 to 2020, offering complementary views of the magnitude, variability, and trends in sea level rise

4.4 Natural Disasters

Stacked Bar Charts: Depicting the frequency of various natural disaster types over time, revealing trends in the prevalence of floods, droughts, and other events (Fig 9).

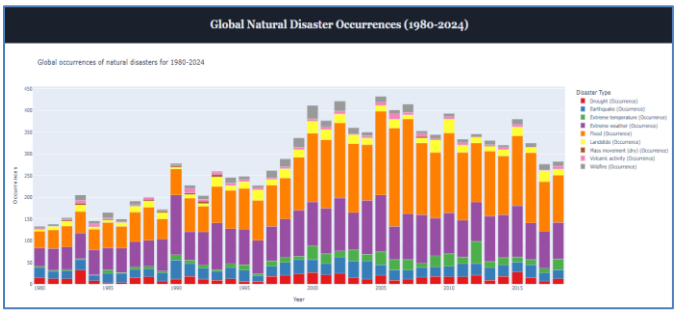

Fig. 9. Global natural disaster occurrences (1980-2024), highlighting trends in disaster frequency

4.5 Correlations

Line Plot (Temperature Anomaly Correlation): The line plot reveals a positive correlation between temperature anomalies and natural disaster occurrences, particularly after 1998, indicating that rising temperatures are linked to increased disaster frequency (Fig 10).

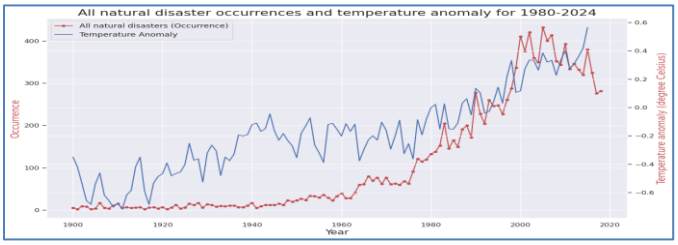

Fig. 10. Correlation between temperature anomalies and natural disasters (1990-2018), showing an increasing trend

Correlation Heatmap: The heatmap shows strong correlations between temperature anomalies, natural disasters, and economic damage, with dry mass movements exhibiting a weaker correlation, while economic damage aligns closely with most disaster types (Fig 11).

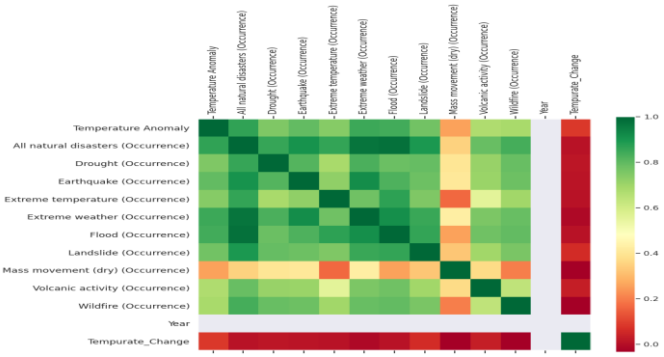

Fig. 11. Heatmap displaying correlations between temperature anomalies and natural disasters

5 Results and Insights

By integrating and visualizing multiple climate-related datasets, we derived key insights:

- **Correlations Between Carbon Emissions and Temperature:** Historical emissions data and surface temperature anomalies reveal a strong positive correlation over time. As global CO_2 emissions increased, notable shifts in temperature anomalies followed. This relationship is most pronounced after the early 20th century, suggesting that anthropogenic activities play a critical role in driving climate change.
- **Rising Sea Levels and Warming:** Sea-level data plotted against global temperature anomalies shows that higher temperatures generally coincide with rising mean sea levels. The temporal analysis indicates that both sea-level rise and temperature increases have accelerated in recent decades, reflecting the compounding effects of ice melt and thermal expansion.
- **Increased Natural Disaster Frequency with Temperature Anomalies**: Our analysis of disaster occurrence data (floods, droughts, extreme weather) in relation to temperature anomalies reveals a positive correlation, especially after the mid-20th century. For instance, floods and extreme weather events have increased markedly as global temperatures deviate from historical norms.
- **Regional Disparities and Long-Term Trends:** Choropleth maps highlight that certain regions have disproportionately high emissions or more pronounced temperature anomalies. Visualizing these spatial variations can guide more targeted climate policies, enabling policymakers to focus mitigation or adaptation measures where they are most needed.

6 Challenges

Challenges Faced:

- **Data Quality and Gaps:** Some datasets contained missing or incomplete values. We applied interpolation and averaging techniques, but these methods introduce uncertainty. Future work should incorporate more robust imputation strategies or leverage supplementary data sources.
- **Data Volume and Complexity:** Handling large, globally distributed datasets posed computational challenges. We addressed these by using efficient data structures and preprocessing steps to standardize units, formats, and temporal resolutions.
- **Complexity of Visualization:** Designing intuitive visualizations for non-linear, multi-dimensional relationships proved challenging. We iterated through several design options to ensure users could easily navigate and interpret the data.

7 FUTURE WORK

Our current platform provides a robust starting point for integrated climate data visualization, but there are several avenues for enhancement:

- **Expanded Data Integration**: Incorporating additional datasets, such as socioeconomic indicators (e.g., GDP, population density), could contextualize climate impacts on human systems. This would enable analyses of climate vulnerability and resilience at finer spatial and demographic resolutions.
- **Forecasting and Predictive Models:** Integrating machine learning techniques could yield predictive models for future temperature trajectories, emissions scenarios, and disaster frequencies. Predictive layers would offer stakeholders a forward-looking tool for planning mitigation and adaptation strategies.

8 CONCLUSIONS

We have presented an interactive, multi-dimensional visualization platform that integrates global climate datasets—including surface temperatures, carbon emissions, sea-level rise, and natural disaster occurrences—into a unified analytical environment. The platform enables users to explore and uncover complex correlations, track long-term trends, and derive insights into how anthropogenic emissions shape climate patterns and disaster risks.

By presenting these diverse datasets side-by-side and in interactive formats, our tool helps reveal relationships that might otherwise remain hidden in isolated analyses. This integrative perspective can inform policymakers, researchers, and the general public, encouraging evidence-based decision-making and proactive climate actions. As data resources grow and predictive models improve, platforms like this can evolve into even more powerful decision-support tools, guiding effective climate policy and fostering resilient communities.

9 SOURCE CODE

https://github.com/CERZ3X/CS524

10 REFERENCES

[1] IPCC AR6 Synthesis Report: Climate Change 2023, Intergovernmental Panel on Climate Change, 2023. Available: https://www.ipcc.ch/report/ar6/syr/.

[2] C. Liu, M. Chen, and H. He, "Interactive climate data analysis dashboard," *IEEE Transactions on Visualization and Computer Graphics*, vol. 25, no. 3, pp. 1150–1162, Mar. 2019.

[3] P. Smith, R. Jones, J. Williams, and K. Brown, "Linking emissions and sea-level rise: A data-driven analysis," *Climate Journal*, vol. 14, no. 2, pp. 45–58, Apr. 2021.

[4] NASA GISS, "Surface temperature analysis," Available: https://data.giss.nasa.gov/gistemp/, accessed 2024.

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